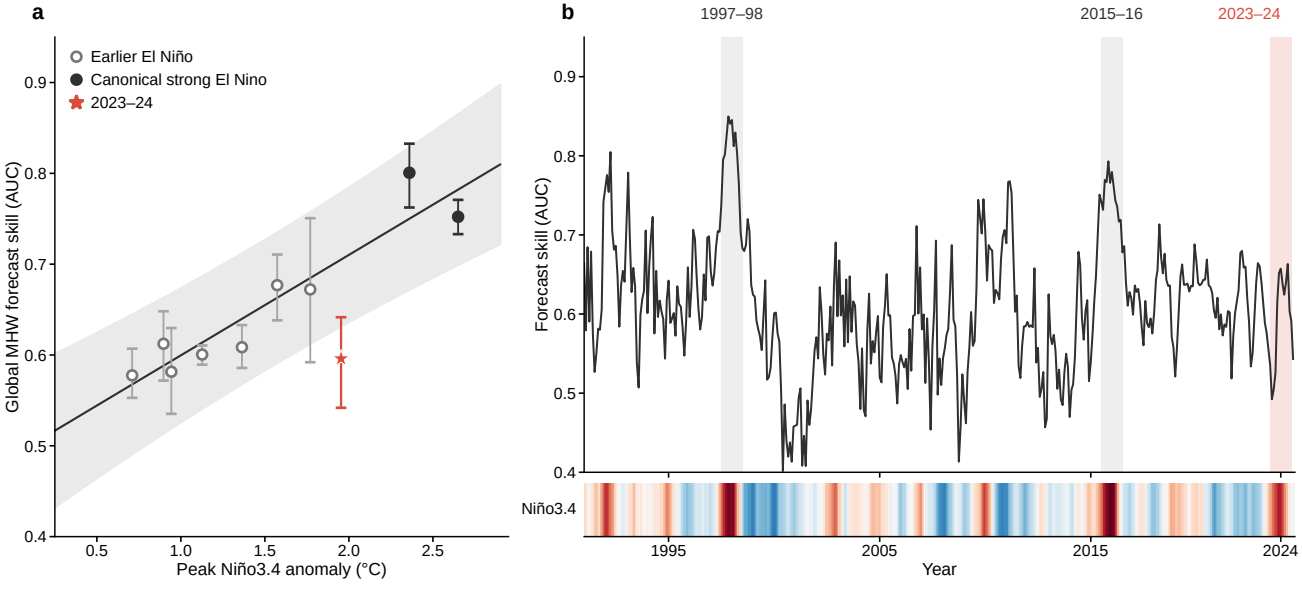
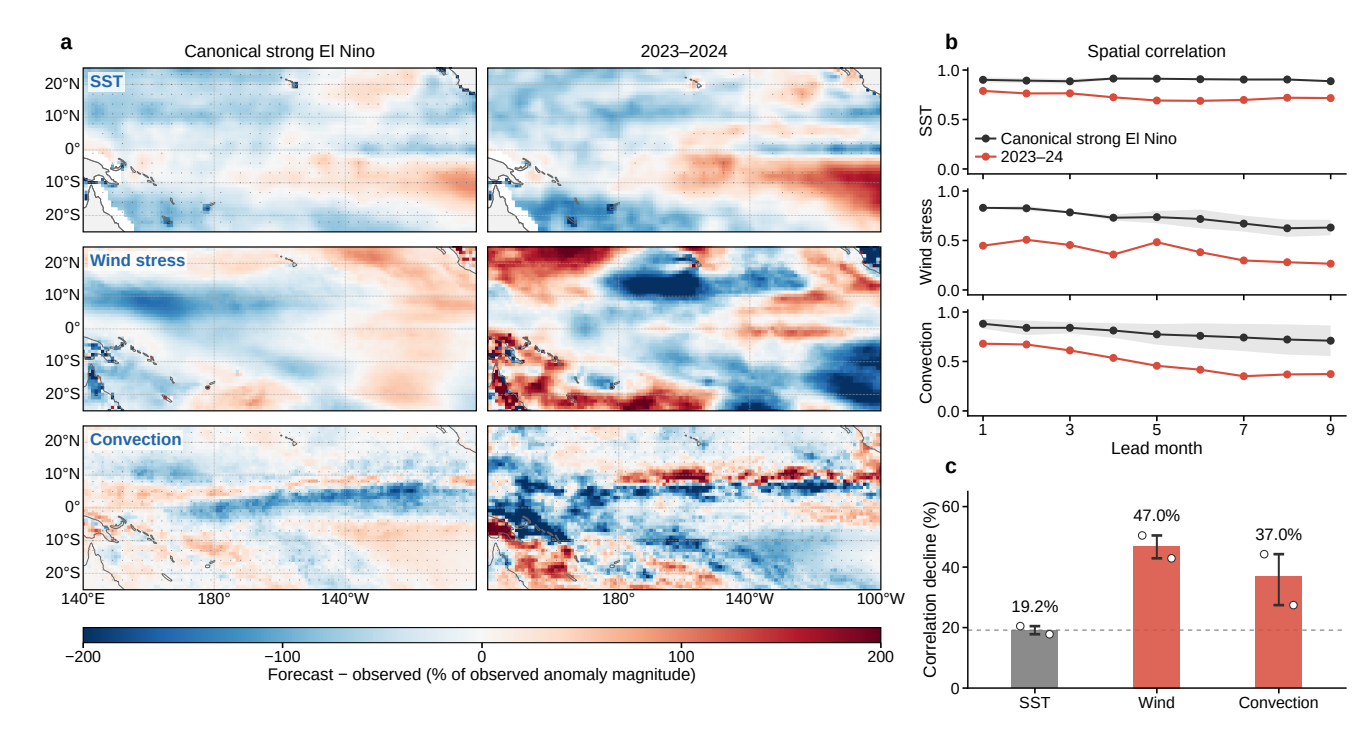
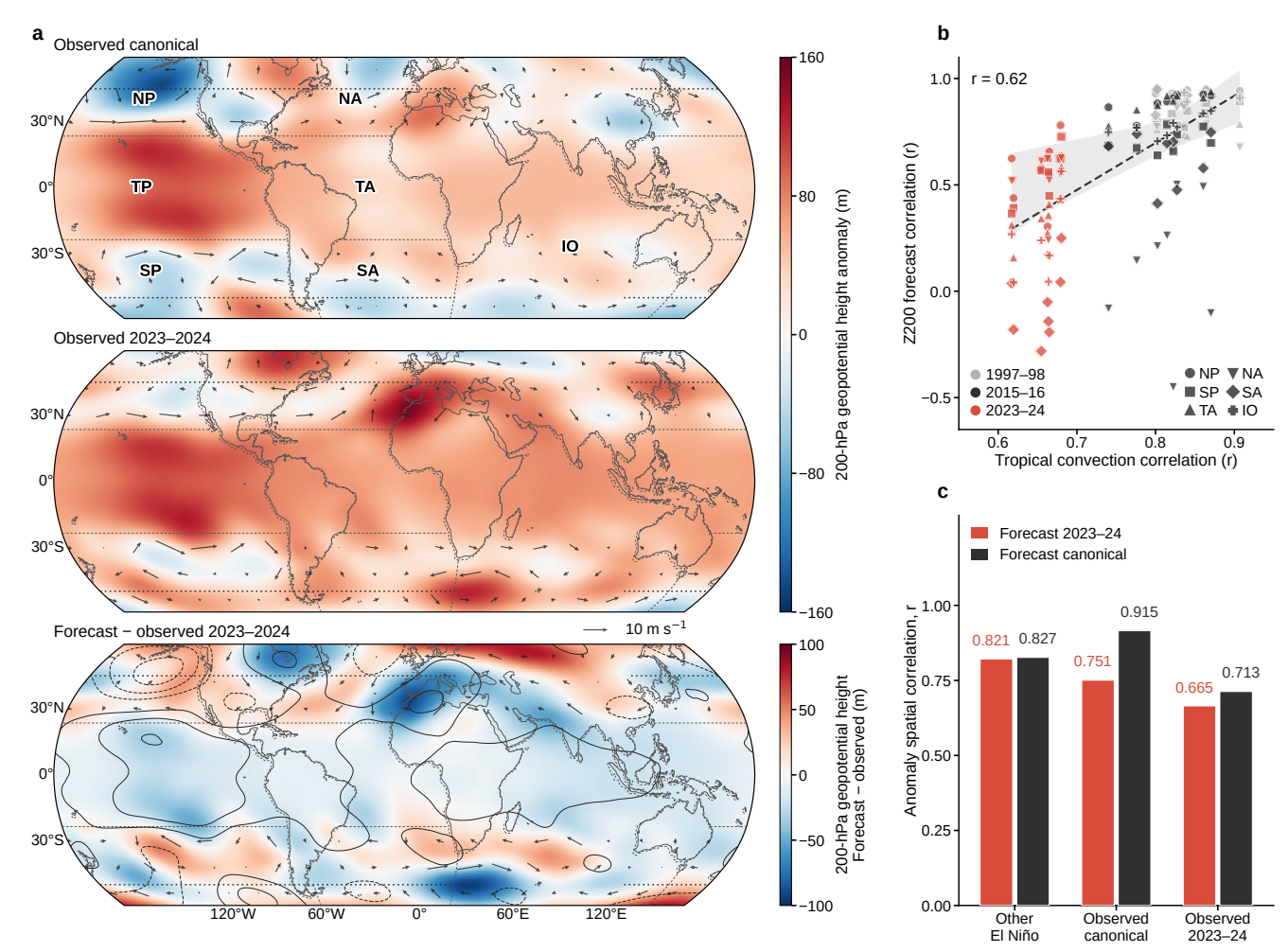
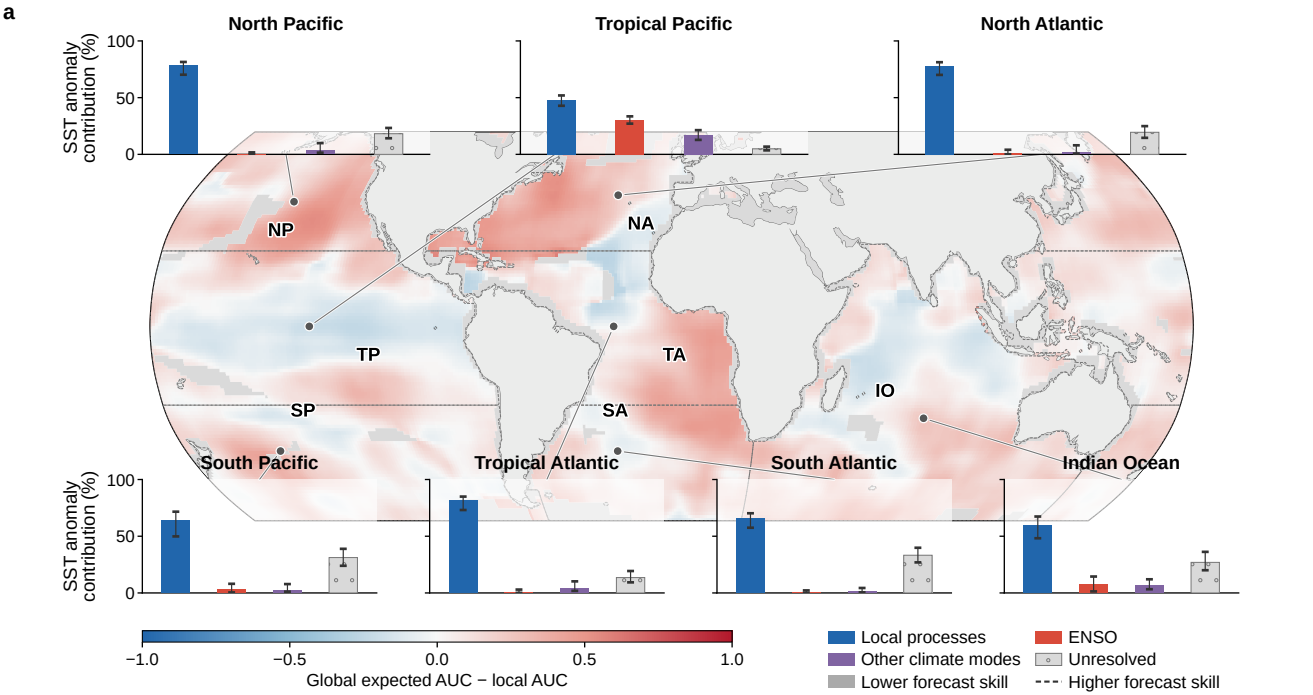
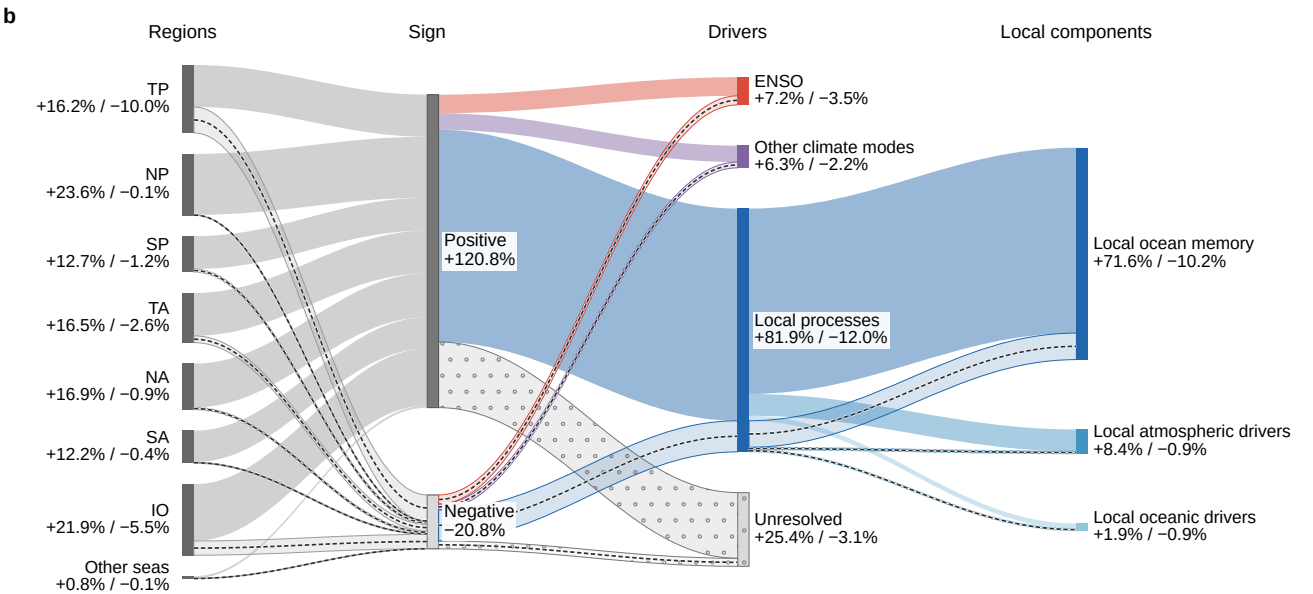

a
Earlier El Niño
Canonical strong El Nino
2023–24
Global MHW forecast skill (AUC)
0.9
0.8
0.7
0.6
0.5
0.4
0.5
1.0
1.5
2.0
2.5
Peak Niño3.4 anomaly (°C)
b
1997–98
2015–16
2023–24
Forecast skill (AUC)
0.9
0.8
0.7
0.6
0.5
0.4
Niño3.4
1995
2005
2015
2024
Year


a
Canonical strong El Nino
2023–2024
SST
Wind stress
Convection
20°N
10°N
0°
10°S
20°S
140°E
180°
140°W
100°W
−200
−100
0
100
200
Forecast − observed (% of observed anomaly magnitude)
b
Spatial correlation
SST
Wind stress
Convection
Canonical strong El Nino
2023–24
1.0
0.5
0.0
1
3
5
7
9
Lead month
c
Correlation decline (%)
0
20
40
60
19.2%
47.0%
37.0%
SST
Wind
Convection


a
Observed canonical
NP
NA
TP
TA
IO
SP
SA
30°N
0°
30°S
Observed 2023–2024
Forecast − observed 2023–2024
10 m s−1
120°W
60°W
0°
60°E
120°E
200-hPa geopotential height anomaly (m)
160
80
0
−80
−160
200-hPa geopotential height Forecast − observed (m)
100
50
0
−50
−100
b
r = 0.62
Z200 forecast correlation (r)
Tropical convection correlation (r)
1997–98
2015–16
2023–24
NP
NA
SP
SA
TA
IO
c
Forecast 2023–24
Forecast canonical
Anomaly spatial correlation, r
0.821
0.827
0.751
0.915
0.665
0.713
Other El Niño
Observed canonical
Observed 2023–24


a
North Pacific
Tropical Pacific
North Atlantic
SST anomaly contribution (%)
100
50
0
NP
NA
TP
TA
SP
SA
IO
South Pacific
Tropical Atlantic
South Atlantic
Indian Ocean
−1.0
−0.5
0.0
0.5
1.0
Global expected AUC − local AUC
Local processes
Other climate modes
Lower forecast skill
ENSO
Unresolved
Higher forecast skill


b
Regions
Sign
Drivers
Local components
TP
+16.2% / −10.0%
NP
+23.6% / −0.1%
SP
+12.7% / −1.2%
TA
+16.5% / −2.6%
NA
+16.9% / −0.9%
SA
+12.2% / −0.4%
IO
+21.9% / −5.5%
Other seas
+0.8% / −0.1%
Positive
+120.8%
Negative
−20.8%
ENSO
+7.2% / −3.5%
Other climate modes
+6.3% / −2.2%
Local processes
+81.9% / −12.0%
Unresolved
+25.4% / −3.1%
Local ocean memory
+71.6% / −10.2%
Local atmospheric drivers
+8.4% / −0.9%
Local oceanic drivers
+1.9% / −0.9%